.4 CO and NO x limits are given on a dry basis. Consequently, if these are measured on a wet basis, those findings should be converted to dry basis reading using a concurrently measured water vapour content in order to

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SOURCE: MEPC.373(80).pdf
.4 CO and NO x limits are given on a dry basis. Consequently, if these are measured on a wet basis, those findings should be converted to dry basis reading using a concurrently measured water vapour content in order to determine the relevant dry/wet correction factor (c oncentration, dry = concentration, wet/exhaust gas non-water fraction);

.5 correction to reference 11.00% O2 should be on the basis of:

Creference = Cmeasured x (20.95 - O2 measured) / (20.95 11.00);

.6 oxygen content, temperature and pressure profiles, as applicable, through the TWTD should be monitored over the duration of each test for conformity with the required values as given by the relevant Functional Objective;

.7 not less than three separate HCl and particulate matter readings should be taken over the duration of each test period at approximately equally spaced intervals and those results averaged to give the result to be compared to the limit value. For thermal waste treatment devices with intermittent loading those test procedures should commence no later than 10 minutes after a loading; and
RESOLUTION MEPC.373(80) (adopted on 7 July 2023)

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SOURCE: MEPC.373(80).pdf
limit value. For thermal waste treatment devices with intermittent loading those test procedures should commence no later than 10 minutes after a loading; and
RESOLUTION MEPC.373(80) (adopted on 7 July 2023)

.8 alternative emission species test methods which provide equivalent results to those given above may be used with the agreement of the Administration.

4.1.3.5 A test report detailing the TWTD tested, the test sequence followed, the measurement devices/procedures used, the traces of the CO, NOx, HC and O2, the temperature and pressure profile readings and the test results of the other emissions measured together with details of the actual sludge oil (if applicable), and solid waste compositions and waste loading quantities and times should form part of the TWTD Technical Report. If the TWTD is not built for handling sludge oil and testing is therefore not undertaken with sludge oil, this should be specified in the test report.

4.1.3.6 In service, the emissions from thermal waste treatment devices certified to

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SOURCE: MEPC.373(80).pdf
sludge oil and testing is therefore not undertaken with sludge oil, this should be specified in the test report.

4.1.3.6 In service, the emissions from thermal waste treatment devices certified to Performance Level 2 should not exceed the following in-service maximum emission limits:

CO 50 ppm (dry basis) at 11.00% O2 averaged over each UTC three-hour period

Opacity 10%

4.1.3.7 CO should be measured in accordance with section 6.4 of the NO x Technical Code 2008 (direct measurement and monitoring) and should be monitored at a frequency of not less than 0.05 Hz.

4.1.3.8 Oxygen content, temperature and pressure profiles, as applicable, through the TWTD should be monitored and controlled in accordance with the relevant Functional Objective.

4.2 Discharge Water to Sea

4.2.1 This section applies if there is:

.1 a direct water discharge as a by -product of the thermal waste treatment process used; and/or

.2 water used to wash the exhaust gas from the TWTD before discharge to the atmosphere, where that water is then subsequently discharged to sea.

Alternatively, these discharge water streams may be collected in a holding tank for

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SOURCE: MEPC.373(80).pdf
.2 water used to wash the exhaust gas from the TWTD before discharge to the atmosphere, where that water is then subsequently discharged to sea.

Alternatively, these discharge water streams may be collected in a holding tank for discharge ashore.

4.2.2 If discharged to sea, the discharge water should not be diluted or mixed with water from other sources before monitoring for the turbidity and PAH limit parameters. After monitoring for PAH and turbidity, the discharge water may be diluted as required or chemically treated prior to pH monitoring.

4.2.3 The discharge water to sea should not exceed the following limits at any time when the TWTD is in operation:

.1 pH: minimum 6.5 or a maximum difference of 2 pH units between the inlet water and the discharged water after dilution values if chemically treated the requirements of 10.1.6.1 of the 2021 Guidelines for Exhaust Gas Cleaning Systems (resolution MEPC.340(77)) should also be applied;

.2 Turbidity: maximum continuous turbidity in the discharge water should not be greater than 25 FNU (formazine nephelometric units) or 25 NTU

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SOURCE: MEPC.373(80).pdf
Cleaning Systems (resolution MEPC.340(77)) should also be applied;

.2 Turbidity: maximum continuous turbidity in the discharge water should not be greater than 25 FNU (formazine nephelometric units) or 25 NTU (nephelometric turbidity units) or equivalent units, above the inlet water turbidity assessed on the basis of 15-minute average values; and

.3 PAH: phenanthrene equivalent concentration should not exceed that equivalent to 2.2 g/h per nameplate capacity in MW at the discharge water flow rate(s) above the inlet water PAH concentration.

4.2.4 The monitoring methods used for pH, turbidity and PAH should be in accordance with the 2021 Guidelines for exhaust gas cleaning systems (resolution MEPC.340(77)) and should be monitored at a frequency of not less than 0.05 Hz.

4.2.5 Performance, calibration and permissible deviations of the discharge water monitoring devices should be in accordance with the relevant sections of the 2021 Guidelines for exhaust

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SOURCE: MEPC.373(80).pdf
4.2.5 Performance, calibration and permissible deviations of the discharge water monitoring devices should be in accordance with the relevant sections of the 2021 Guidelines for exhaust gas cleaning systems (resolution MEPC.340(77)).

4.3 Residues from thermal waste treatment devices

4.3.1 Any solid residues or other materials from TWTD, including any washings or other material collected as part of maintenance or servicing activities, should be discharged ashore to appropriate reception facilities.

4.3.2 Any residues from a TWTD discharge water treatment system, either in-service or as collected during maintenance or servicing activities, should be discharged ashore to appropriate reception facilities.

5 Functional Objectives and TWTD Technical Report

5.1 These are the Functional Objectives which should be met in order to achieve the in-service Performance Level 1 emission limit requirements and, if applicable, those of Performance Level 2. The following listing of core Functional Objectives represents a technology-neutral approach to the review of the design, manufacture, installation, use and

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SOURCE: MEPC.373(80).pdf
Performance Level 2. The following listing of core Functional Objectives represents a technology-neutral approach to the review of the design, manufacture, installation, use and ongoing management of a TWTD. The applicant is therefore additionally responsible for identifying any other Functional Objectives which may potentially affect the device's performance in terms of emissions to air and, if applicable, water and to duly address those as part of the TWTD Technical Report such that the requirements of regulation 4.4 of MARPOL Annex VI are met. Consequently, the TWTD Technical Report is to cover, but is not limited to, an assessment of the following functional requirements and is to be compiled against the Functional Objective references as listed in table 2. In the case of operational, servicing or maintenance requirements, the TWTD Tech nical Report may cite the relevant section of the TWTD Operating Manual which is to be supplied with the device rather than reproducing in full the applicable text. Where a particular Functional Objective is not applicable owing to the operating principle applied, the waste streams to be processed or other factors would be given

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SOURCE: MEPC.373(80).pdf
the applicable text. Where a particular Functional Objective is not applicable owing to the operating principle applied, the waste streams to be processed or other factors would be given as not applicable, together with supporting justification, in the TWTD Technical Report.